



This product is to be used only by a registered medical practitioner with experience in cardiology

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contain

Amiodarone (INN) hydrochloride.....200mg

PHARMACEUTICAL FORM

White, round tablets, scored and enengraved with a “heart-shaped” symbol and “200” on one face.

CLINICAL PARTICULARS

Therapeutic indications

Cordarone is indicated only for the treatment of severe rhythm disorders, not responding to other therapy or when other treatments cannot be used:

- atrial rhythm disorders (conversion of fibrillation or flutter, and maintenance of sinus rhythm following conversion)
- nodal rhythm disorders
- ventricular rhythm disorders (life threatening ventricular premature contractions, non-sustained ventricular tachycardia, prevention of ventricular tachycardia or ventricular fibrillation episodes.
- rhythm disorders, associated with Wolf-Parkinson-White syndrome

In view of its pharmaceutical properties, amiodarone is indicated in the above rhythm disorders especially when they are associated with an underlying heart disease (coronary insufficiency, heart failure).

Posology and method of administration

Initial treatment

The usual dosage regimen is 3 tablets daily for 8 to 10 days.

In some cases, higher doses have been used for initial treatment (4 to 5 tablets daily), always for short periods and under electrocardiographic monitoring.

Maintenance treatment

The minimum effective dose, which varies depending on the patient, should be determined, and ranges from half a tablet daily (1 tablet every 2 days) to 2 tablets daily.

Pediatric population

The safety and efficacy of amiodarone in children have not been established. See Sections *pharmacodynamics properties* and *pharmacokinetics properties* for currently available data.

Method of administration

Oral route.

Contraindications

This medicinal product is contraindicated in the following cases:

- sinus bradycardia and sinoatrial block in patients without a pacemaker;
- sick sinus syndrome in patients without a pacemaker (risk of sinus arrest);
- high-degree atrioventricular conduction disorders in patients without a pacemaker;
- hyperthyroidism because of possible exacerbation by amiodarone;
- known hypersensitivity to iodine, amiodarone or to one of the excipients;
- 2nd and 3rd trimesters of pregnancy;
- breast-feeding;
- in combination with:
 - medicinal products that may induce torsades de pointes (excluding antiparasitic drugs, neuroleptics and methadone):
 - o class Ia antiarrhythmics (quinidine, hydroquinidine, disopyramide),
 - o class III antiarrhythmics (sotalol, dofetilide, ibutilide),
 - o other medicinal products, such as arsenic compounds, bepridil, cisapride, citalopram, escitalopram, diphemanil, dolasetron IV, domperidone, dronedarone, erythromycin IV, levofloxacin, mequitazine, mizolastine, vincamine IV, moxifloxacin, prucalopride, spiramycin IV, toremifene (see *Interaction with other medicinal products and other forms of interaction*).
 - telaprevir,
 - cobicistat.

Special warnings and precautions for use

Special warnings

Cardiac disorders:

An ECG must be performed before starting treatment. Slowing of heart rate may be accentuated in elderly patients.

Amiodarone induces ECG changes. This cordarone-induced change consists of QT interval prolongation as a result of prolonged repolarization, with possible appearance of a U wave. This is a sign of therapeutic effect and not of toxicity.

The occurrence during treatment of 2nd or 3rd degree AV block, sinoatrial block or bifascicular block requires treatment discontinuation. First degree AV block requires increased monitoring. The onset of a new arrhythmia or the worsening of pre-existing and treated arrhythmia has been reported (see *Undesirable effects*).

Such a proarrhythmic effect may occur especially if there are factors that promote QT interval prolongation, such as certain drug combinations and/or hypokalemia (see Sections *Interaction with other medicinal products and other forms of interaction* and *Undesirable effects*). The risk of drug-induced torsades de pointes seems lower with amiodarone than with other anti-arrhythmic agents causing the same degree of QT interval prolongation.

Thyroid disorders

This medicinal product contains iodine and therefore interferes with some thyroid tests (radioactive iodine binding, PBI). Thyroid function tests (T3, T4, usTSH) can however be performed.

Amiodarone can cause thyroid disorders, particularly in patients with a history of thyroid dysfunction. An assay of TSH is recommended in all patients before starting treatment, then regularly throughout treatment and several months after treatment discontinuation, and if there is clinical suspicion of dysthyroidism (see *Undesirable effects*).

Pulmonary disorders

The onset of dyspnea or a dry cough, alone or associated with a deterioration in the general condition, should suggest the possibility of pulmonary toxicity, such as interstitial pneumonitis, and requires radiological assessment (See *Undesirable effects*).

Liver disorders

Regular monitoring of liver function is recommended at the start of treatment, then regularly throughout treatment with amiodarone (see *Undesirable effects*).

Neuromuscular disorders

Amiodarone can cause sensory, motor or mixed peripheral neuropathy and myopathy (see *Undesirable effects*)

Eye disorders

If blurred vision or deterioration in visual acuity occur, a full ophthalmological examination, including funduscopy, should be promptly performed. Neuropathy or optic neuritis, induced by amiodarone, requires treatment discontinuation due to the potential risk of progression to blindness (see *Undesirable effects*).

Severe skin disorders

Life-threatening or even fatal cutaneous reactions such as Stevens-Johnson syndrome or toxic epidermal necrolysis may occur. Should signs or symptoms indicative of these conditions (e.g. progressive skin rash with blisters or mucosal lesions) occur, amiodarone treatment should be discontinued immediately.

Severe bradycardia and conduction disorders

Cases of severe, potentially life-threatening bradycardia and conduction disorders have been observed when amiodarone is used in combination with sofosbuvir in combination with another antiviral (DAA) acting directly on the hepatitis C virus (HCV), such as daclatasvir, simeprevir, or ledipasvir.

Bradycardia has usually occurred within a few hours to a few days, but cases with a longer time to onset have been observed, mostly up to 2 weeks after initiation of anti-HCV treatment.

Amiodarone should only be used in patients treated with medicines containing sofosbuvir if they are intolerant or contraindicated to other antiarrhythmic drugs.

If concomitant use of amiodarone is deemed necessary, it is recommended that patients undergo inpatient cardiac monitoring for the first 48 hours of co-administration, following which outpatient monitoring or heart rate self-monitoring should be performed daily for at least the first 2 weeks of treatment.

In view of the long half-life of amiodarone, cardiac monitoring as described above should also be performed in patients who have stopped amiodarone within the last few months and who are to start treatment with medicines containing sofosbuvir alone or in combination with other DAAs.

All patients who currently or have recently used amiodarone in combination with medicines containing sofosbuvir should be warned of the symptoms of bradycardia and conduction disturbances, and they should be advised of the need for urgent medical attention if they experience these symptoms.

Transplantation

In retrospective studies, amiodarone use in the transplant recipient prior to heart transplant has been associated with an increased risk of primary graft dysfunction (PGD).

PGD is a life-threatening complication of heart transplantation that presents as left, right or biventricular dysfunction occurring within the first 24 hours of transplant surgery for which there is no identifiable secondary cause (see *Undesirable effects*). Severe PGD may be irreversible.

For patients who are on the heart transplant waiting list, consideration should be given to use an alternative antiarrhythmic drug as early as possible before transplant.

Drug interactions:

Combination (see *Interaction with other medicinal products and other forms of interaction*) with:

- beta-blockers other than sotalol (contraindicated combination) and esmolol (combination requiring precautions for use)
- verapamil and diltiazem should only be considered for the prevention of life-threatening ventricular arrhythmias.

Concomitant administration of amiodarone with the following medicinal products is not recommended: ciclosporin, diltiazem (by injection) or verapamil (by injection), certain antiparasitics (halofantrine, lumefantrine and pentamidine), certain neuroleptics (amisulpride, chlorpromazine, cyamemazine, droperidol, flupentixol, fluphenazine, haloperidol, levomepromazine, pimozide, pipamperone, pipotiazine, sertindole, sulpiride, sultopride, tiapride, zuclopenthixol), fluoroquinolones (other than levofloxacin and moxifloxacin), stimulant laxatives, methadone or fingolimod (see *Interaction with other medicinal products and other forms of interaction*).

Excipients:

This medicinal product contains lactose. It is not recommended in patients with galactose intolerance, Lapp lactase deficiency or glucose galactose malabsorption syndrome (rare hereditary diseases).

Precautions for use

Electrolyte disturbances, particularly hypokalemia: it is important to take into account situations liable to be associated with hypokalemia, which may favor the onset of proarrhythmic effects.

Hypokalemia should be corrected before amiodarone is administered.

The adverse effects mentioned below are most often related to drug overload; they can be avoided or minimized by carefully determining the minimum maintenance dose.

Patients are advised to avoid sun exposure or to take protective measures against the sun during treatment.

The safety and efficacy of amiodarone in children has not been evaluated in controlled clinical studies.

Due to the possible increase in the defibrillation and/or pacing threshold of implantable cardiac defibrillators or pacemakers, the threshold should be checked before and several times after amiodarone treatment initiation and whenever dosage is adjusted.

Anesthesia

Before surgery, the anesthesiologist should be informed that the patient is on amiodarone.

The adverse effect of chronic amiodarone therapy are likely to add to the hemodynamic risk associated with general or local anesthesia. These effects include in particular bradycardia, hypotension, reduced cardiac output and conduction disorders.

Furthermore, some cases of acute respiratory distress syndrome have been observed immediately after surgery in patients treated with amiodarone. These patients should be closely monitored when mechanically ventilated (see *Undesirable effects*).

Interaction with other medicinal products and other forms of interaction

Antiarrhythmic agents

Many antiarrhythmics have depressant effects on cardiac automaticity, conduction and contractility.

Concomitant use of antiarrhythmics from different classes can be beneficial, but this therapeutic approach often proves problematic and requires ECG and close clinical monitoring. Concomitant use of antiarrhythmics that may induce torsades de pointes

(amiodarone, disopyramide, quinidine compounds, sotalol, etc.) is contraindicated. Concomitant use of antiarrhythmics from the same class is not recommended except in exceptional cases, due to the increased risk of cardiac adverse effects. Concomitant use with medicinal products that have negative inotropic properties, that induce bradycardia and/or slow atrioventricular conduction is problematic, and requires clinical and ECG monitoring.

Medicinal products that may induce torsades de pointes

This serious arrhythmia can be induced by a number of medicinal products, regardless of whether they are antiarrhythmics. Hypokalemia (see “Potassium-lowering agents”) is a predisposing factor, as is bradycardia (see “Bradycardic agents”) or a congenital or acquired pre-existing prolongation in QT interval.

These medicinal products include class Ia and III antiarrhythmics and certain neuroleptics. For dolasetron, erythromycin, spiramycin and vincamine, this interaction only occurs with IV forms. In general, using two torsadogenic drugs concomitantly is contraindicated. However, this does not apply to some of these agents which are considered absolutely necessary and, instead of being contraindicated, are simply not recommended in combination with other torsadogenic medicinal products. This concerns methadone, antiparasitics (halofantrine, lumefantrine, pentamidine) and neuroleptics.

Bradycardia inducing agents

Numerous medicinal products can induce bradycardia. This is the case particularly with class Ia antiarrhythmic drugs, beta blockers, some class III antiarrhythmic drugs, some calcium antagonists, digitalis drugs, pilocarpine and anticholinesterase agents.

Effect of amiodarone on other medicinal products

Amiodarone and/or its metabolite, desethylamiodarone, inhibit CYP1A1, CYP1A2, CYP3A4, CYP2C9, CYP2D6 and P-glycoprotein and may increase exposure of their substrates.

Given the long-acting effect of amiodarone, these interactions may be observed for several months after treatment discontinuation.

Effect of other medicinal products on amiodarone

CYP3A4 inhibitors and CYP2C8 inhibitors may potentially inhibit amiodarone metabolism and therefore increase exposure.

CYP3A4 inhibitors (e.g. grapefruit juice and certain medicinal products) should preferably not be used during amiodarone treatment.

Contraindicated combinations

+ Medicinal products that may induce torsades de pointes (except for antiparasitics, neuroleptics and methadone; see “Inadvisable combinations”):

- class Ia antiarrhythmics (quinidine, hydroquinidine, disopyramide),
- class III antiarrhythmics (dofetilide, ibutilide, sotalol),
- other medicinal products, such as: arsenic compounds, bepridil, cisapride, citalopram, escitalopram, diphemanil, dolasetron IV, domperidone, dronedarone, erythromycin IV, levofloxacin, mequitazine, mizolastine, vincamine IV, moxifloxacin, prucalopride, spiramycin IV, toremifene.

Increased risk of ventricular arrhythmias, particularly torsades de pointes.

+ Telaprevir

Cardiac automaticity and conduction disorders with risk of excessive bradycardia.

+ Cobicistat

Risk of increased amiodarone-induced adverse effects due to decreased metabolism.

Inadvisable combinations

+ Sofosbuvir

Co-administration of amiodarone with treatments containing sofosbuvir may result in severe symptomatic bradycardia. Use only if no alternative treatment is available. Close monitoring is recommended when these medicinal products are co-administered (see section **Special warnings and precautions for use**).

+ CYP3A4 substrates

Amiodarone is an inhibitor of CYP3A4 and increases plasma concentrations of CYP3A4 substrates, potentially increasing the toxicity of these substrates.

+ Ciclosporin

Increase in circulating ciclosporin levels due to reduced metabolism by the liver, with the risk of nephrotoxic effects.

Assay of serum ciclosporin concentrations, monitoring of renal function and adjustment of the ciclosporin dose during treatment with amiodarone.

+ Diltiazem injection

Risk of bradycardia and atrioventricular block. If this combination cannot be avoided, close clinical supervision and continuous ECG monitoring should be performed.

+ Fingolimod

Potential of the bradycardia-inducing effects with potentially fatal outcome. This is particularly true for beta-blockers which inhibit adrenergic compensation mechanisms. Clinical supervision and continuous ECG monitoring for 24 hours following the first dose should be performed.

+ Verapamil injection

Risk of bradycardia and atrioventricular block. If this combination cannot be avoided, close clinical supervision and continuous ECG monitoring should be performed.

+ Antiparasitics that may induce torsades de pointes (halofantrine, lumefantrine, pentamidine)

Increased risk of ventricular arrhythmias, particularly torsades de pointes. If possible, discontinue 1 of the 2 treatments. If this combination cannot be avoided, monitor QT before instituting treatment and monitor ECG.

+ Neuroleptics that may induce torsades de pointes (amisulpride, chlorpromazine, cyamemazine, droperidol, flupentixol, fluphenazine, haloperidol, levomepromazine, pimozide, pipamperone, pipotiazine, sertindole, sulpiride, sultopride, tiapride, zuclopenthixol).

Increased risk of ventricular arrhythmias, particularly torsades de pointes.

+ Methadone

Increased risk of ventricular rhythm disorders, especially torsades de pointes.

+ Fluoroquinolones other than levofloxacin and moxifloxacin (contraindicated combinations):

Increased risk of ventricular rhythm disorders, especially torsades de pointes.

+ Stimulant laxatives

Increased risk of ventricular arrhythmias, particularly torsades de pointes (hypokalemia is a predisposing factor).

Correct any hypokalemia before administering the medicinal product and carry out ECG and clinical monitoring, together with electrolyte monitoring.

+ Fidaxomicine

Increased plasma fidaxomicine concentrations.

Combinations requiring precautions for use

+ P-glycoprotein substrates

Amiodarone is a P-glycoprotein (P-gp) inhibitor. Co-administration with P-gp substrates may lead to increased exposure of these substrates.

+ Digitalis drugs

Suppressed automaticity (excessive bradycardia) and atrioventricular conduction disorders. If digoxin is used, blood digoxin levels can be increased due to reduced digoxin clearance, requiring ECG and clinical monitoring.

If necessary, blood digoxin levels should be monitored and the digoxin dose adjusted.

+ Dabigatran

Increase in plasma dabigatran concentrations, with a higher risk of bleeding.

If dabigatran is used postoperatively, clinical monitoring should be performed and, if necessary, the dabigatran dose should be adjusted, without exceeding 150 mg/day.

+ CYP2C9 substrates

Amiodarone increases plasma concentrations of CYP2C9 substrates such as vitamin K antagonists and phenytoin.

+ Oral anticoagulants (Vitamin K antagonists)

Increase in the anticoagulant effect and in the risk of bleeding.

More frequent control of INR. Possible dose adjustment of the oral anticoagulant during treatment with amiodarone and for 8 days after treatment discontinuation.

+ Phenytoin (and, by extrapolation, fosphenytoin)

Increase in plasma phenytoin concentrations with signs of overdose, particularly neurological signs (decreased liver metabolism of phenytoin).

Clinical monitoring and monitoring of plasma phenytoin concentrations should be performed and, if necessary, the phenytoin dose should be adjusted.

+ CYP2D6 substrates

• Flecainide

Amiodarone increases plasma concentrations of flecainide by inhibiting cytochrome CYP2D6. The flecainide dose should be adjusted.

+ CYP3A4 substrates

Amiodarone is an inhibitor of CYP3A4 and increases plasma concentrations of substrates of this cytochrome, potentially increasing the toxicity of these substrates.

• Statins (simvastatin, atorvastatin, lovastatin)

The risk of muscle toxicity (e.g. rhabdomyolysis) is increased by concomitant administration of amiodarone as statins can be metabolized by CYP3A4. Use of another statin not affected by this type of interaction is recommended.

+ Other drugs metabolized by CYP3A4 (lidocaine, tacrolimus, sildenafil, midazolam, dihydroergotamine, ergotamine, colchicine, triazolam)

Amiodarone is an inhibitor of CYP3A4 and increases plasma concentrations of these molecules, potentially increasing the toxicity of these substances.

+ Lidocaine

Risk of increased lidocaine plasma concentrations, potentially leading to neurological and cardiac adverse effects, due to decreased liver metabolism by amiodarone.

Clinical and ECG monitoring and, if necessary, monitoring of plasma lidocaine concentrations should be performed. If necessary, the lidocaine dose should be adjusted during treatment and after amiodarone discontinuation.

+ Tacrolimus

Increase in blood tacrolimus concentrations due to inhibition of its metabolism by amiodarone.

Assay of blood tacrolimus concentrations, monitoring of renal function and tacrolimus dose adjustment should be performed during combined treatment with amiodarone and after amiodarone discontinuation.

+ Beta-blockers other than sotalol (contraindicated combination) and esmolol (combination requiring precautions for use)

Automaticity and conduction disorders (suppressed compensatory sympathetic mechanisms). ECG and clinical monitoring should be performed.

+ Beta-blockers in heart failure (bisoprolol, carvedilol, metoprolol, nebivolol)

Automaticity and cardiac conduction disorders with risk of excessive bradycardia.

Increased risk of ventricular arrhythmias, particularly torsades de pointes. Regular clinical and ECG monitoring should be performed.

+ Esmolol

Contractility, automaticity and conduction disorders (suppressed compensatory sympathetic mechanisms).

ECG and clinical monitoring should be performed

+ Oral diltiazem

Risk of bradycardia or atrioventricular block, particularly in the elderly.

ECG and clinical monitoring should be performed.

+ Certain macrolides (azithromycin, clarithromycin, roxithromycin)

Increased risk of ventricular rhythm disorders, especially torsades de pointes.

ECG and clinical monitoring should be performed during combined treatment with amiodarone.

+ Oral verapamil

Risk of bradycardia and atrioventricular block, particularly in the elderly. ECG and clinical monitoring should be performed.

+ Hypokalemic agents: hypokalemic diuretics (alone or in combination), amphotericin B (IV route), glucocorticoids (systemic route), tetracosactide

Increased risk of ventricular arrhythmias, particularly torsades de pointes (hypokalemia is a predisposing factor).

Hypokalemia should be corrected before the medicinal product is administered and ECG, electrolyte and clinical monitoring should be performed.

+ Bradycardic agents

Increased risk of ventricular arrhythmias, particularly torsades de pointes.

ECG and clinical monitoring should be performed.

+ Orlistat

Risk of a reduction in plasma concentrations of amiodarone and of its active metabolite.

Clinical monitoring and, if necessary, ECG monitoring should be performed.

+ Tamsulosin

Risk of increased tamsulosin-induced adverse effects due to inhibition of its hepatic metabolism.

Clinical monitoring should be performed and the tamsulosin dose adjusted during treatment with the enzyme inhibitor and after its discontinuation, if necessary.

+ Voriconazole

Increased risk of ventricular arrhythmias, particularly torsades de pointes, as amiodarone metabolism may be decreased.
Clinical and ECG monitoring should be performed and the amiodarone dose adjusted if necessary.

Combinations to be taken into account

+ Pilocarpine

Risk of excessive bradycardia (additive effects of bradycardic agents).

Fertility, pregnancy and lactation

Pregnancy

Animal studies have not demonstrated any teratogenic effects. In the absence of a teratogenic effect in animals, no teratogenic effects are expected in humans.

To date, substances causing malformations in humans have been shown to be teratogenic in animals during well-conducted studies in two species.

In a clinical context, there are not yet sufficient relevant data in order to evaluate the possible teratogenic effect of amiodarone when administered during the first trimester of pregnancy.

Since the fetal thyroid gland begins to bind iodine from week 14 of pregnancy, no effects on the fetal thyroid gland are expected in the event of administration before this timepoint. Iodine overload with the use of this drug beyond this period may give rise to biological or clinical (goiter) fetal hypothyroidism.

Consequently, the use of this medicinal product is contraindicated from the 2nd trimester of pregnancy.

Lactation

Amiodarone and its metabolite, together with iodine, are excreted in breast milk at concentrations greater than those in maternal plasma. Due to the risk of hypothyroidism in the newborn infant, breast-feeding is contraindicated during treatment with this medicinal product.

Undesirable effects

The undesirable effects have been classified by system-organ and by incidence as follows:
Very common ($\geq 10\%$); common ($\geq 1\%$, $< 10\%$); uncommon ($\geq 0.1\%$, $< 1\%$); rare ($\geq 0.01\%$, $< 0.1\%$); very rare ($< 0.01\%$); not known (cannot be estimated from the available data).

Eye disorders:

Very common: Corneal microdeposits, in almost all adults, usually limited to the area under the pupil and not requiring discontinuation of amiodarone. Exceptionally, they are associated with colored haloes in dazzling light, or blurred vision. Corneal microdeposits consist of complex lipid deposits, and are always completely reversible following treatment discontinuation.

Very rare: Optic neuropathy (optic neuritis) with blurred and decreased vision and funduscopy showing papillary edema, possibly progressing to more or less severe decrease in visual acuity. A causal relationship with amiodarone has not currently been established. However, if there is no other clear cause, it is recommended to discontinue amiodarone treatment.

Skin and subcutaneous tissue disorders:

Very common: Photosensitivity. It is recommended to avoid exposure to sun (and ultraviolet rays in general) during treatment.

Common: Lilac or slate grey colored skin pigmentations, occurring with high daily dosages as prolonged treatment, slowing disappearing after treatment discontinuation (10 to 24 months).

Very rare: Erythema during radiotherapy. Skin rashes, usually non-specific. Exfoliative

dermatitis, although a causal relationship with the drug is not clearly established. Alopecia.
Not known: Eczema, severe, sometimes fatal, cutaneous reactions such as toxic epidermal necrolysis (Lyell's syndrome) and Stevens-Johnson syndrome, bullous dermatitis, DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms).

Endocrine disorders:

• Thyroid disorders.

Very common: Unless there are clinical signs of thyroid dysfunction, "unrelated" blood thyroid hormone changes (increased T4, normal or slightly reduced T3) do not warrant treatment discontinuation.

Common: Hypothyroidism causes usual symptoms: weight gain, cold intolerance, apathy, drowsiness. A marked increase in TSH levels confirms the diagnosis. After treatment discontinuation, normal thyroid function is gradually restored within 1 to 3 months. Treatment discontinuation is not essential: if justified by the indication, treatment with amiodarone can be continued in combination with thyroid hormone replacement therapy using levothyroxine. Levothyroxine doses can be adjusted according to TSH levels.

Hyperthyroidism is more difficult to identify: symptoms are fewer (slight, unexplained weight loss, reduced efficacy of anti-angina and/or antiarrhythmic therapy); psychiatric symptoms in elderly patients, even thyrotoxicosis.

A marked drop in ultrasensitive TSH levels confirms the diagnosis. It is essential to discontinue amiodarone treatment, which is usually sufficient to trigger clinical recovery within 3 to 4 weeks. As serious cases can be fatal, urgent appropriate treatment should be instituted. When thyrotoxicosis is a cause for concern, either directly or due to its effects on the fragile myocardial balance, the variable efficacy of synthetic antithyroid drugs makes it necessary to recommend high-dose corticosteroids (1 mg/kg) over a sufficiently long period (3 months). Cases of hyperthyroidism have been reported up to several months following amiodarone discontinuation.

• Other endocrine disorders

Very rare cases of SIADH (syndrome of inappropriate antidiuretic hormone secretion), particularly when used concomitantly with medicinal products that can induce hyponatremia. See also "Investigations".

Respiratory, thoracic and mediastinal disorders:

Common: Cases of interstitial or diffuse alveolar pneumonitis and bronchiolitis obliterans with organizing pneumonia (BOOP), sometimes fatal, have been reported. The onset of exertional dyspnea or a dry cough, alone or associated with a deterioration in general health (fatigue, weight loss and slight fever) requires radiological assessment and, if necessary, treatment discontinuation. These types of pneumonitis can lead to pulmonary fibrosis.

Early withdrawal of amiodarone treatment, with or without corticosteroid therapy, leads to regression of symptoms. Clinical signs usually resolve in 3 or 4 weeks; radiological and lung function improvement is slower (several months).

Some cases of pleurisy, usually associated with interstitial pneumonitis, have been reported.

Very rare: Bronchospasm, particularly in asthmatic patients.

Acute respiratory distress syndrome, occasionally with a fatal outcome, sometimes in the period immediately after surgery (a possible interaction with high doses of oxygen has been suggested) (see *Special warnings and special precautions for use*).

Not Known (cannot be estimated from the available data): Cases of pulmonary bleeding that can manifest in some cases as hemoptysis have been reported. These pulmonary effects are often associated with amiodarone-induced pneumonitis.

Nervous system disorders:

Common: Tremor or other extra-pyramidal symptoms. Sleep disorders including nightmares. Peripheral sensorimotor or mixed neuropathy.

Uncommon: Myopathy. Peripheral sensorimotor or mixed neuropathy or myopathy may occur after just a few months of treatment but sometimes after several years. They are usually

reversible on treatment discontinuation. However, recovery may be incomplete, very slow and only be observed several months after stopping treatment.

Very rare: Cerebellar ataxia. Benign intracranial hypertension, headache. If occasional headaches occur, investigations for possible underlying disease must be carried out.

Not known: Syndrome parkinsonism, parosmia.

Hepatobiliary disorders:

Cases of liver damage, diagnosed based on elevated serum transaminases, have been reported as follows:

Very common: Generally moderate, isolated elevation in transaminases (1.5 to 3 times normal range) resolving after dose reduction, or even spontaneously.

Common: Acute liver damage with elevation in blood transaminases and/or jaundice, sometimes with a fatal outcome, requiring treatment discontinuation.

Very rare: Chronic liver damage during prolonged treatment. Histological findings are consistent with pseudoalcoholic hepatitis. As clinical and laboratory signs are not explicit (variable hepatomegaly, elevation in blood transaminases between 1.5 and 5 times normal range) regular monitoring of liver function is warranted.

Chronic hepatic damage should be suspected in case of elevation, even moderate, in blood transaminases, occurring after treatment lasting more than 6 months. The clinical and biological disorders usually regress after treatment discontinuation. A few irreversible cases have been reported.

Cardiac disorders:

Common: Bradycardia, generally moderate and dose-dependent.

Uncommon: Conduction disorders (sinoatrial block, various degrees of AV block)

Very rare: Marked bradycardia and, more exceptionally, sinus arrest, reported in certain cases (sinus dysfunction, elderly patients).

Not known: Torsades de pointes (see sections *Special warnings and precautions for use* & *Interaction with other medicinal products and other forms of interaction*)

Gastrointestinal disorders:

Very common: Minor digestive disorders (nausea, vomiting, dysgeusia), which generally occur during initial treatment and disappear when dosage is reduced.

Not known: Pancreatitis/acute pancreatitis, dry mouth, constipation.

Breast and reproductive system disorders:

Very rare: Epididymitis. A causal relationship with the medicinal product has not been clearly established.

Vascular disorders:

Very rare: Vasculitis.

Investigations:

Rare: Rare cases of hyponatremia, suggestive of SIADH.

Very rare: Renal impairment with increased serum creatinine.

Blood and lymphatic system disorders:

Very rare: Thrombocytopenia, haemolytic anaemia, aplastic anaemia.

Not known: Neutropenia, agranulocytosis

Immune system disorders:

Not known: Cases of angioedema and/or urticaria have been reported. Anaphylactic/anaphylactoid reaction, and even shock.

General disorders:

Not known: Granuloma, essentially bone marrow granuloma, have been reported.

Metabolism and nutrition disorders

Not known: Decreased appetite.

Psychiatric disorders

Common: Libido decreased

Not known: Confusional state, delirium, hallucination.

Musculoskeletal and connective tissue disorders:

Not known: Lupus syndrome.

Injury, poisoning and procedural complications

Not known:

Potentially fatal primary graft dysfunction post cardiac transplant (see *Special warnings and special precautions for use*).

Overdose

There is little information available regarding acute overdose with oral amiodarone. A few cases of sinus bradycardia, ventricular arrhythmias, particularly torsades de pointes, and hepatic impairment have been reported. Treatment must be symptomatic. Due to the pharmacokinetics of the drug, prolonged surveillance of the patient, particularly cardiac status, is recommended.

Amiodarone and its metabolites are not dialyzable.

PHARMACOLOGICAL PROPERTIES**Pharmacodynamic properties**

Pharmacotherapeutic group: ANTIARRHYTHMICS, CLASS III

ATC code: C01BD01

Antiarrhythmic properties:

- Prolongation of phase 3 of the action potential of cardiac fibers essentially caused by a reduction in the potassium current (Vaughan Williams class III).
- Bradycardiac effect by reducing sinus automaticity. This effect is not antagonized by atropine;
- Non-competitive alpha and beta anti-adrenergic effect;
- Slowed sinoatrial, atrial and nodal conduction, which is more pronounced the faster the rhythm;
- No changes in intraventricular conduction;
- Increase in refractory periods and reduction in myocardial excitability on an atrial, nodal and ventricular level;
- Slowing of conduction and prolongation of refractory periods in the atrioventricular accessory pathways.

Other properties:

- Reduction in oxygen consumption due to a moderate decrease in peripheral resistance and reduced heart rate;
- Increased coronary blood flow due to a direct effect on smooth muscle of myocardial vessels and sustained cardiac output through reduced pressure and peripheral resistance and the absence of negative inotropic effects.

A meta-analysis was performed of 13 prospective, randomized, controlled studies including 6 553 patients who had recently experienced myocardial infarction (78%) or chronic heart failure (22%).

Mean follow-up of patients ranged from 0.4 to 2.5 years. Mean daily maintenance dose ranged MY/COR200mg/1025/FRSmPC0625

from 200 to 400 mg.

This meta-analysis demonstrated that amiodarone significantly reduced overall mortality by 13% (95% CI: 0.78 - 0.99; $p = 0.030$) and rhythm-related mortality by 29% (95 % CI: 0.59 - 0.85; $p = 0.0003$).

However, these results must be interpreted cautiously due to the heterogeneity of the various studies (differences related mainly to the included population, duration of follow-up, methodology used and study results).

The percentage of patients discontinuing treatment was higher in the amiodarone group (41%) than in the placebo group (27%).

Seven percent of the patients taking amiodarone developed hypothyroidism, *versus* 1% in the placebo group. Hyperthyroidism was diagnosed in 1.4% of patients taking amiodarone, *versus* 0.5% in the placebo group.

Interstitial pneumonitis occurred in 1.6% of patients taking amiodarone, *versus* 0.5% in the placebo group.

Pediatric population

No controlled clinical studies have been conducted in children. In the published literature, the safety of amiodarone has been studied in 1118 children with various types of arrhythmia.

The following doses were used in pediatric clinical studies:

- loading dose: 10 to 20 mg/kg/day for 7 to 10 days (i.e. 500 mg/m²/day if dose is expressed as BSA).
- maintenance dose: the minimum effective dose should be used; based on individual response, this can range from 5 to 10 mg/kg/day (i.e. 250 mg/m²/day if dose is expressed as BSA).

Pharmacokinetic properties

Amiodarone is a slow transport compound with a high tissue affinity.

Its bioavailability by oral route varies depending on individual patients from 30% to 80% (mean value 50%). After a single dose, peak plasma concentrations are reached in 3 to 7 hours. Therapeutic activity is obtained, on average, within one week (from a few days to two weeks). The half-life of amiodarone is long, with marked inter-individual variability (20 to 100 days). During the first days of treatment, the medicine accumulates in most body tissues, particularly in the adipose tissue. Elimination begins after a few days and the input/output ratio balances out after one to several months, depending on the patient.

These characteristics warrant the use of loading doses to rapidly achieve the level of tissue uptake required for therapeutic activity.

Some iodine is released and is recovered in the urine as iodide; i.e. 6 mg/24 hours for a daily dose of 200 mg of amiodarone. The rest of the compound, and therefore most of the iodine, is excreted in the feces after hepatic transport.

Since negligible amounts of the drug are eliminated in the urine, usual dosages can be used in patients with kidney failure.

After treatment discontinuation, elimination continues for several months. It should be noted that residual activity can occur for 10 days to 1 month.

Amiodarone is mainly metabolized by cytochrome CYP3A4, and also by cytochrome CYP2C8. Amiodarone and its metabolite, desethylamiodarone, are potential *in vitro* inhibitors of cytochromes CYP1A1, CYP1A2, CYP2C9, CYP2C19, CYP2D6, CYP3A4, CYP2A6, CYP2B6 and CYP2C8. Amiodarone and desethylamiodarone can also inhibit transport proteins such as P-gp and organic cation transporter 2 (OCT2). One study showed a 1.1% increase in creatinine concentration (an OCT2 substrate).

In vivo data describe an interaction between amiodarone and CYP3A4, CYP2C9, CYP2D6 and P-gp substrates.

Pediatric population

No controlled clinical studies have been conducted in children.

Available literature data, which are limited, show no difference in pharmacokinetic parameters between adults and children.

Preclinical safety data

In a 2-year carcinogenicity study in rats, amiodarone caused an increase in the number of thyroid follicular tumors (adenomas and/or carcinomas) in both sexes at clinically relevant exposures.

Since mutagenicity findings were negative, an epigenetic rather than genotoxic mechanism has been suggested to explain induction of this type of tumor.

Studies in mice did not show any carcinomas, but dose-dependent thyroid follicular hyperplasia was observed. These effects on the thyroid in rats and mice were probably due to the effects of amiodarone on the synthesis and/or release of thyroid hormones. These findings have little relevance to humans.

PHARMACEUTICAL PARTICULARS**Shelf life**

Do not use later than the date of expiry.

How supplied

30 tablets, packed in blister.

Storage Condition

Store below 30°C.

Manufacturer

Sanofi Winthrop Industrie
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Date of Revision: October 2025